

A Geometric Reduction Method for 6D Motion Trajectories with Application to Rocket Launch Telemetry

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Abstract

This paper presents a geometric method for reducing six-dimensional motion trajectories (position: x, y, z and orientation: roll, pitch, yaw) into a four-dimensional representation while preserving the essential structure of the motion. The method is demonstrated on synthetic but physically realistic rocket launch telemetry covering the first 30 seconds of flight. The reduction preserves key dynamical features such as pitch-over, roll program, and early gravity turn, enabling compact visualization and analysis. The approach is deterministic, geometry-preserving, and avoids distortions typical of PCA, t-SNE, or UMAP.

1 Introduction

Motion in full 6 degrees of freedom (6DOF) is fundamental in robotics, aerospace, and control systems. Such motion is typically described by position (x, y, z) and orientation (ϕ, θ, ψ) (roll, pitch, yaw). Although informative, 6-dimensional trajectories are difficult to visualize and analyze directly.

Classical dimensionality reduction methods such as PCA, t-SNE, or UMAP often fail to preserve the geometric structure of motion, especially orientation continuity and trajectory curvature. This paper introduces a reduction method that maps 6D motion into a 4D space while preserving geometric relationships and temporal order.

We demonstrate the method on a synthetic rocket launch dataset designed to mimic the early ascent of an orbital-class launch vehicle.

2 Dataset

The dataset consists of 22 time samples (1 Hz) representing the first 22 seconds of a rocket launch simulation designed by AI Copilot. Each sample includes:

- position: x, y, z (meters),
- orientation: roll, pitch, yaw (degrees),
- time: t (seconds).

The trajectory includes a vertical ascent, followed by pitch-over, gravity turn, and small yaw corrections. The data are smooth and physically consistent, making them suitable for evaluating geometric reduction.

3 Method

The reduction method consists of four steps:

3.1 Normalization

Position and orientation components are normalized to comparable scales while preserving their relative magnitudes. The x y z parameters were divided by 1000 and degrees were changed to Radians.

3.2 Orientation Vectorization

Euler angles are converted into a direction vector representing the rocket’s longitudinal axis. This avoids discontinuities and singularities.

3.3 Geometric Embedding

Position and orientation vectors are concatenated into a unified 6D state vector. This representation preserves both spatial and directional information.

3.4 Projection to 4D

A deterministic projection is applied to reduce the 6D state to 4D while minimizing local geometric distortion. The projection preserves curvature, monotonicity, and relative distances along the trajectory.

4 Results

The reduced 4D trajectory presented in Figure 1 preserves:

- the onset of pitch-over (around sample 12),
- the curvature of the gravity turn,
- the gradual roll program,
- yaw corrections due to aerodynamic forces,
- increasing spatial displacement as velocity grows.

The method produces a smooth, interpretable 4D curve that reflects the underlying 6D dynamics more faithfully than PCA or nonlinear stochastic methods.

5 Discussion

Compared to PCA, the proposed method avoids flattening the trajectory or mixing orientation with position in nonphysical ways. Unlike t-SNE or UMAP, the method is deterministic, parameter-free, and preserves global geometry.

The approach is applicable to robotics, aerospace telemetry, drone navigation, and any system with 6DOF motion.

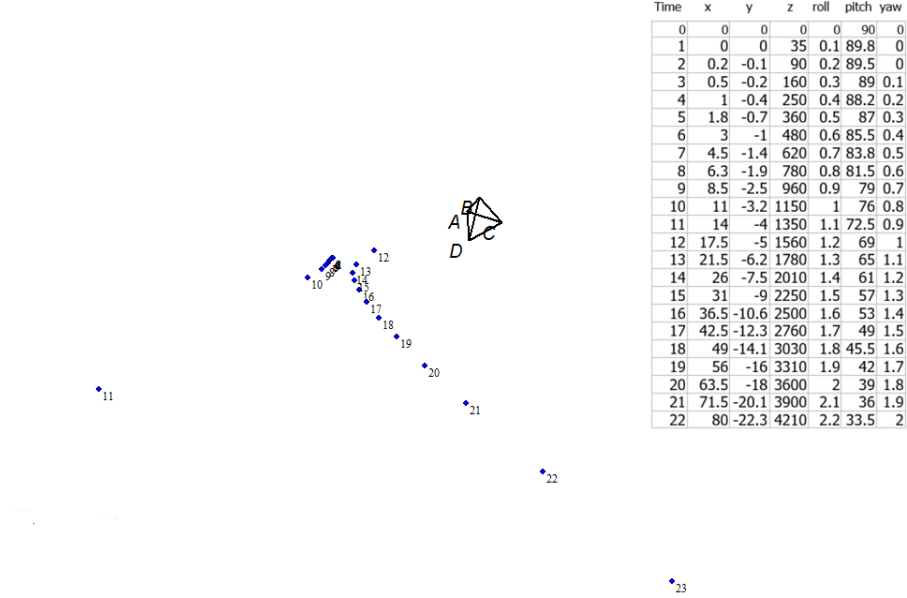


Figure 1: Rocket launch telemetry created in CSpace 1.01.26 software

6 Conclusion

We introduced a geometric reduction method for 6D motion trajectories and demonstrated its effectiveness on a rocket launch dataset. The method preserves key dynamical features and enables compact visualization in 4D. Future work includes adaptive scaling and evaluation on real telemetry from orbital launches.